AURA

AI Virtual DJ

User Manual

Plan and perform beat-matched DJ sets from your own music, with mid-song mix points, a library of transition effects and DDJ-FLX4 style performance controls. Everything runs in your browser (or the Android app). Your audio never leaves your device.

Contents

Section	What it covers
1. Quick start	Load music, build a set, press play
2. The screen	Where everything lives
3. Library	Adding, selecting, reordering and removing tracks
4. Building a set	How the AI DJ plans order, exits, entries and effects
5. Decks and timelines	Playing, seeking, the master timeline
6. Mixer strip and pads	EQ, CFX, Smart CFX, hot cues, loops, jumps, pad FX
7. Beat FX and Loop effect	Tempo-synced performance effects
8. Transition effects reference	All 15 transition styles
9. Shortcuts	Keyboard and mouse
10. Android app	Building and using the APK
11. Troubleshooting and limits	Common issues and what is not included

1. Quick start

- **Add music.** Drag audio files or a whole folder onto the Library (or anywhere on the page), or press **Files** / **Folder** in the Sources card. No music handy? Press **Demo** to synthesise 12 practice tracks.
- **Wait for analysis.** Each track shows "analysing..." and then its detected BPM. Analysis happens in your browser.
- **Choose effects.** Leave all effect chips on and let the DJ decide, or switch off the ones you do not want (or press **Clean blends only**).
- **Set the length.** Use the **Tracks in set** slider (3 to 10).
- **Press Build set.** The Set order list, the master timeline and DJ's notes fill in with the plan.
- **Click a vinyl** on Deck A or Deck B to start the set. Click it again to pause.

Tip. You need at least two analysed tracks to build a set. Large files can take several seconds to analyse. Wait for the BPM to appear before pressing Build set.

2. The screen

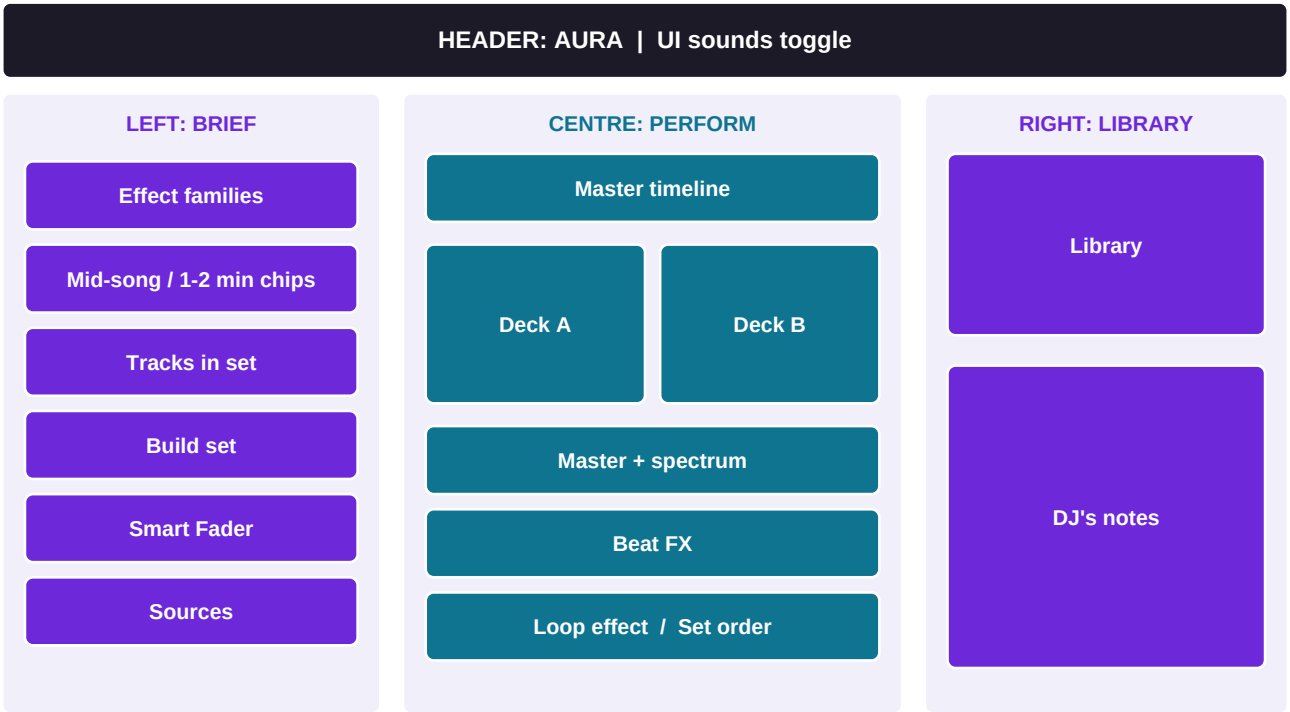

Figure 1 - Screen layout (desktop). On narrow screens the three columns stack vertically.

Area	Purpose
Left: Brief the DJ	Effect families, entry/exit options, set length, Build set, Smart Fader, Sources
Centre	Master timeline, Deck A and Deck B (with mixer strips and pads), master meter, Beat FX, Loop effect, Set order
Right	Library and DJ's notes (a running log of every decision the DJ makes)
Header	AURA title and the UI sounds toggle (short clicks when you press controls)

3. Library

The Library holds every track you have loaded. Rows show the name, detected BPM, a guessed genre label (taken from the file name) and an energy bar.

Adding tracks

- **Drag and drop** files or folders onto the Library, or anywhere on the page. Folders are read including sub-folders.
- **Files** opens a file picker; **Folder** opens a folder picker (desktop browsers only).
- Supported formats are whatever your browser can decode: normally MP3, WAV, M4A/AAC, FLAC and OGG.

Selecting and removing

- Click a row (or its checkbox) to select it. **Shift-click** selects a range. **Select all** is at the top of the list.
- **Remove (n)** or the **Delete** key removes the selected tracks; the **X** on a row removes just that track.
- **Undo** appears after a removal and restores the tracks to their original positions.
- Tracks in the set that is currently playing cannot be removed. Stop the set first. Removing tracks from a planned set clears the plan so you can rebuild it.

Reordering

Drag a row onto another row to move it. If several rows are selected, dragging one of them moves the whole group. Library order does not affect the AI's set order; it is for your own organisation.

Privacy

Audio is decoded and analysed locally. Nothing is uploaded.

Tip. For a quicker library, name files with hints such as "bhangra", "phonk" or "90s". AURA uses them for the genre label.

4. Building a set

Pressing **Build set** runs the AI DJ. It does four jobs.

4.1 Analysing each track

AURA estimates tempo (about 79 to 176 BPM), finds the beat grid, and measures loudness bar by bar. From that it knows where each track has intros, breakdowns, builds and drops, and every candidate mix point sits on a bar boundary so the beats stay locked.

4.2 Choosing the order

The DJ follows an energy arc: a Party arc that starts moderate, climbs to a peak about 70 percent of the way through, then eases. It opens with a track close to the starting energy and then picks each next track to keep the tempo step small (it also considers half and double time) while following the arc.

4.3 Choosing exit and entry points

- **Exit (out) point** can be anywhere on the outgoing track's timeline, not just the end. It prefers phrase boundaries, energy that suits the next move, and a sensible amount of play time.
- **Entry (in) point** can be anywhere on the incoming track: its intro, its drop, or a high-energy phrase in the middle. The set list labels each as "(from start)" or "(mid-song)".
- Switch off the **Mid-song entry points** chip to force every entry from the start.
- Switch on **Exit 1-2 min for songs played from the start** to make the DJ exit any track that began at 0:00 between 1:00 and 2:00 (if a track is too short for that window, the DJ falls back to its normal choice). You can toggle this during playback and the next exit is re-planned immediately.

4.4 Choosing effects

Every allowed effect is scored against the tempo gap, the change in energy, the target energy at that point of the set, and what was used recently. The DJ avoids repeating an effect or an effect family back to back, limits the big dramatic moves (riser, gater, tape stop, reverse swell, loop roll) to one every few transitions, and is more conservative on the first transition. Wide tempo gaps (above about 8 percent) favour cuts: echo-out, tape stop or reverse swell. Rising energy favours riser or gater builds. Falling energy favours reverb, phaser or filter. Similar tempo and energy favours sweeps, swirls and drive.

4.5 Reading the plan

- **Set order** shows each track, its BPM change, an effect tag (hover for the reason) and "out m:ss, in m:ss".
- **DJ's notes** logs the energy and BPM path, the full transition plan, and a sentence for every transition when it happens.
- **Master timeline** shows the whole set as coloured segments with the next effect under each.

Tip. Change the effect chips or the 1-2 minute option and press Build set again to get a different plan. The DJ is deterministic: the same library and settings give the same plan.

5. Decks and timelines

Play and pause

Click the **vinyl** on either deck. The glyph shows what a click will do (play or pause). It controls both decks together. If no set is built yet, the first click builds one and starts it. Space bar does the same.

Deck timeline (YouTube style)

- Click or drag the bar to jump. Hover to preview the time. The bar thickens and shows a knob as you hover.
- Click the waveform to jump to that point.
- **-10s / +10s** buttons skip back and forward.
- A **cyan flag** marks where the AI plans to mix out; the label beside the time shows the next effect and exit time.
- Seeking is locked while a transition is running, so a blend cannot be broken mid-way.

Deck readout

Each deck shows the track name, current BPM (after tempo matching), genre label, pitch change in percent, and elapsed / total time. During a transition the incoming deck lights up as it takes over.

Master timeline

- A bar across the top of the centre column shows the whole set. Each segment is one track's on-air stretch; the label underneath names the effect leading to the next track.
- The white playhead moves as the set plays.
- **Drag or click** anywhere on it to jump to that moment in the set. The right track starts at the right position.
- **Restart set** jumps back to 0:00.

Smart Fader

The **Smart Fader - mix next now** button starts the planned transition immediately, using the planned effect and entry point, but timed to the next beat. Use it to skip ahead when you do not want to wait for the planned exit.

6. Mixer strip and performance pads

Each deck has a mixer strip and pads modelled on a two-channel controller such as the Pioneer DDJ-FLX4. They work on any track you loaded, and they act on that deck whenever it is playing.

Mixer strip

Control	Range	What it does
HI / MID / LOW	-26 to +6 dB (bottom = kill)	3-band EQ at 3.5 kHz, 1 kHz and 200 Hz. Double-click to reset to 0.
CFX	-1 to +1	Colour filter. Left of centre is a low-pass sweep; right of centre is a high-pass sweep. Double-click to centre.
VOL	0 to 1	Channel fader.
Smart CFX	4 presets + off	Press to cycle Dub Echo, Reverb, Crush, Drive, then off. Turning CFX adds filter plus that effect, and fades the track out at the extremes for a smooth exit.

Pad modes

Choose a mode with the tabs under the strip. There are eight pads per deck.

Mode	Pads	How to use
Hot Cue	8 cue pads	Click an empty pad while the track plays to set a cue at the current position. Click a set pad to jump there. Shift-click to clear.
Pad FX	Echo, Flanger, Reverb, Repeater, Phaser, Gater, Crush, Drive	Hold to play the effect on that deck; release to let it ring out. Repeater loops half a beat while held.
Beat Loop	1/4, 1/2, 1, 2, 4, 8, 16, 32 beats	Click to loop that many beats from the last beat. Click the same pad again to exit.
Beat Jump	-1, +1, -2, +2, -4, +4, -8, +8 beats	Click to jump the playhead by that many beats.

Tip. Pads and the mixer strip only affect the playing deck, and effect pads pause while a transition is running because the AI is using those effects itself. Smart CFX and the CFX knob reset when a transition starts.

7. Beat FX and Loop effect

Beat FX

A tempo-synced effect unit, like the Beat FX section of a controller. Pick an effect, a beat length and a channel, set the level, then switch FX ON. It stays latched until you switch it off.

Setting	Options
Effect	Echo, Flanger, Phaser, Chorus, Reverb, Autopan, Crush, Drive, Gater
Beats	1/4, 1/2, 1, 2, 4 (echo delay time, modulation cycle length, or gate speed)
Channel	Deck A, Deck B or Both
Level	10 to 100 percent wet amount (gate depth for Gater)

Changing any setting while FX is on re-applies it immediately. Latched effects are re-applied automatically after each transition.

Loop effect (1 to 5 seconds)

- Set the **Length** slider from 1 to 5 seconds (steps of 0.5).
- Press **Loop live deck** to loop that length, starting from the most recent beat. Press again (**Exit loop**) to carry on.
- Moving the slider while a loop is active changes the loop length at once.
- A loop exits on its own after 16 seconds so it cannot block the planned transition. The AI also exits any loop automatically when it is time to mix.

Tip. Try a 1 second loop on a build, then release it right on a bar line for an instant drop.

8. Transition effects reference

These are the 15 transition styles the DJ can choose from. They are grouped into six families. Turn any chip off in the brief to stop the DJ using it; a clean blend is always available.

Family	Effect	What you hear
Clean	Blend	Beat-matched crossfade over 16 beats with a bass swap at the halfway point.
Delay and echo	Echo-out	Outgoing track is cut on a downbeat and trails off in a dark, ping-pong dub echo while the new track hits. Best for wide tempo gaps.
Modulation	Flanger sweep	Jet-style swoosh on the outgoing track during the blend.
	Phaser sweep	Softer swirling notches, good for cool-downs.
	Chorus swirl	Shimmering thickening of the outgoing track.
	Autopan sweep	Outgoing track pans left and right in time as it fades.
Filter and build	Filter sweep	High-pass sweep on the outgoing track while the new one opens up from muffled.
	Riser and drop	Accelerating snare roll and noise riser, then a sub boom as the new track drops in on the beat with a sidechain-style pump.
	Gater build	Outgoing track stutters on and off at 1/8, 1/16 and 1/32 notes, then the drop.
Space	Reverb wash	Outgoing track dissolves into a reverb tail as the new one fades in.
	Reverse swell	The last bar of the outgoing track plays backwards through reverb, then the new track hits.
Distortion and lo-fi	Overdrive push	Outgoing track is driven harder as it fades under the new one.
	Bitcrush fade	Outgoing track degrades into lo-fi crunch while it fades.
Cut, brake and loop	Tape stop	Outgoing track slows to a halt like a turntable brake, then the new track enters.
	Loop roll	Repeater on the outgoing track: a 1-beat loop shrinks to 1/2, 1/4 and 1/8 before the drop.

Blend length depends on the energy arc: the plan uses 16 beats for the default Party arc. Short cuts (echo-out, tape stop, loop roll) have fixed lengths.

9. Shortcuts

Input	Action
Space	Play / pause (same as clicking a vinyl)
Left / Right arrow	Seek the live deck -5 / +5 seconds
J / L	Seek the live deck -10 / +10 seconds
Delete	Remove selected library tracks
Shift-click (library)	Select a range of tracks
Shift-click (hot cue pad)	Clear that cue
Double-click (EQ / CFX)	Reset the control to centre
Click / drag (timelines)	Jump to that position
Drag (library row)	Reorder tracks; drop files on the Library to add them

10. Android app

AURA can be wrapped as an Android app (APK). A ready-made project is supplied as **aura-dj-android-project.zip**.

- Create a GitHub repository and upload the project contents (including the hidden **.github** folder).
- Open the **Actions** tab and run **Build AURA APK**. After a few minutes download the **AURA-DJ-debug-apk** artifact.
- Copy **app-debug.apk** to your phone, allow "Install unknown apps", and tap it.
- The APK is signed for testing. Publishing to Google Play needs your own signing key.

On Android use **Files** to add music (folder picking and drag-and-drop are not available). Tap once to start audio. Some controls are small on phone screens.

11. Troubleshooting and limits

Problem	What to try
No sound	Check device volume, then click a vinyl once so the browser allows audio. Make sure the tab is not muted.
Build set does nothing	You need at least two analysed tracks. Wait for BPM labels to appear. Clear any failed ("error") tracks from the library.
Mix sounds out of sync	Detection can pick half or double tempo on songs with live drums or tempo changes. Check the BPM shown; use Smart Fader or Restart set, and tell the developer the song and BPM.
A track shows "error"	The browser could not decode the file. Convert to MP3 or WAV.
Seek does not work	Seeking is locked during a transition and only works on the live deck.
A loop will not exit	Press Exit loop, or wait: loops end automatically after 16 seconds.
Effects missing from the plan	You turned them off in the brief, or the DJ judged other effects a better fit. Press All on and rebuild.
No planned mid-song entry	The Mid-song chip is off, or the track had no clear sections (very even energy).

What AURA does not include (yet)

- Streaming services such as Spotify. Their audio cannot be accessed by a web page, so only your own files are supported.
- Musical key detection and harmonic mixing.
- Sampler and keyboard pad modes, and a manual tempo slider / sync button.
- Genre or mood selection (the DJ always plans a Party arc), and genre detection from audio (only file-name hints).
- Recording or exporting the mix.

Tips for the best sets

- Use full songs with clear drums rather than very ambient material; beat detection works best on steady beats.
- Load 8 to 20 tracks so the DJ has real choices, and build a set of 5 to 8.
- Keep all effects on for variety, then remove any you dislike.
- Use headphones and keep the browser tab active; background tabs slow the scheduler.

AURA is a creative tool. Only use music you have the right to play.